

MILGRAM'S EXPERIMENT

(six degrees of separation)

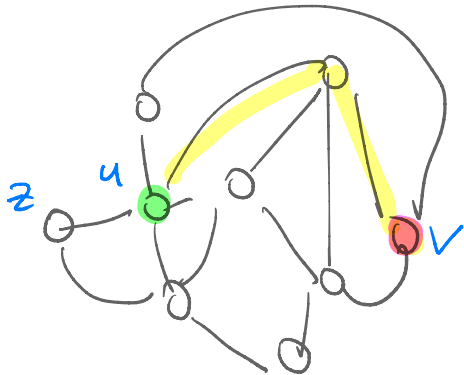
Friendship network : $G=(V,E)$
(Acquaintance)

$V = \{ \text{people} \}$

$E = \{ xy : x, y \in V \text{ and } x \text{ and } y \text{ know each other} \}$

Two randomly chosen nodes u, v :

$d(u, v)$ = the length of the shortest path connecting u and v



$$d(u, v) = 2$$

$$d(z, v) = 3$$

In Milgram's terminology
 u and v have $d(u, v) - 1$
intermediaries

Milgram asked for the average (distance - 1) in our terminology

He selected 296 volunteers, three groups:

A: 100 people from Boston

B: 100 people from Nebraska who were stakeholders ↖ to get a feature

C: 96 people from Nebraska

Target T was a stakeholder from Boston, so that T was in the class of A and B

He asked people from $A \cup B \cup C$ to send a letter to T:

the letter could be given from x to y only if they knew each other

Outcome: only 64 letters were successfully delivered to T

The average number of intermediaries was 5.2

$\Rightarrow$ avg. distance was 6.2 $\Rightarrow$ this motivated the term
Six degrees of separation

(some birds)

In modern times, the experiment was repeated using Facebook.

(NYT 2011)

the avg was 3.74

$\Rightarrow$ the avg distance was 4.74

in Feb 2011

Let's address here the computational task in a simplified way.

Distance distribution

$$\text{For any given } h \geq 0, \quad N_h = \frac{|\{ (u,v) \in V \times V : d(u,v) = h \}|}{n(n-1)}$$

is the fraction of ordered node pairs
at distance exactly h

$\rightarrow$ ordered pairs
as the network
could be directed

$$\text{Average distance} = \sum_{u,v \in V} \frac{d(u,v)}{n(n-1)} = \sum_{h \geq 0} h N_h$$

$\rightarrow$ can be seen as
the weighted sum
on the distribution

If we know N_h , we can compute
average distance ("almost" Milgram's goal)

What we do: we approximate N_w , hence we get an estimation of the average distance

Use random sampling on the nodes of V
(here randomization is in the choice of the nodes)

Recall that $d(u, v)$ can be computed by a BFS on unweighted graphs or by Dijkstra's shortest path algorithm on weighted graphs.

Choose randomly $U \subseteq V$, a subset of nodes from V

Define $N_h^U = \frac{|\{ (u,v) \in U \times V : d(u,v) = h \}|}{|U| (n-1)}$

Setting $U = V$, we get back to N_h

For a small set U , we obtain $\tilde{N}_h^U \approx N_h^U$

(in other word, we perform BFS/Dijkstra starting just from the randomly chosen nodes, i.e., from U)

Running time goes from $O(|V|^2)$ to $O(\tilde{U} |V|)$
 $\uparrow$
 $g(|V|)$ if Dijkstra no good if $\tilde{U} = O(g(|V|))$

Using Azuma-Hoeffding bound (a variant of CB)

we can get a FPTAS (ϵ, δ) when $|U| = O(\epsilon^{-2} \lg S^{-1})$

Hint: $X_i = N_h^{\{u_i\}}$ is fraction, so $X_i \in [0, 1]$ but it is

$U = \{u_1, u_2, \dots, u_k\}$, $|U| = k$ no more an integer
 $a_i \leq X_i \leq b_i$

$$N_h^U = \frac{\sum_{i=1}^k N_h^{\{u_i\}}}{k} = \frac{\sum_{i=1}^k X_i}{k} \Rightarrow E[N_h^U] = \frac{1}{k} \sum_{i=1}^k E[N_h^{\{u_i\}}] = \frac{k}{k} N_h$$

$$E[X_i] = E[N_h^{\{u_i\}}] = \frac{1}{|U|} \sum_{z \in U} N_h^{\{z\}} = N_h^V = N_h$$

Applying AH:

$$\mu = N_h \quad X_i = N_h^{a_i} \quad a_i = 0, b_i = 1$$

$$\Pr \left[\left| \frac{\sum_{i=1}^n X_i}{n} - \mu \right| \geq \varepsilon \right] \leq 2^{-\frac{2h^2\varepsilon^2}{\sum_{i=1}^n (b_i - a_i)^2}} = 2^{-2h\varepsilon^2} < \delta$$

$$\Pr \left[\left| \underbrace{N_h^0}_{\tilde{N}_h} - N_h \right| \geq \varepsilon \right]$$

$$\Downarrow \\ h = O(\varepsilon^{-2} \log \delta^{-1})$$

□